higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

INSTRUMENT TECHNOLOGY
NQF LEVEL 2

18 NOVEMBER 2019

This marking guideline consists of 8 pages.

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QUESTION 1

1.1 1.1.1 Process instrumentation refers to a collection of instruments✓ that we use in process plants to observe,✓ measure and control the variables of a process.✓ (3)

1.1.2
- Pressure
- Flow
- Level
- Temperature (4)

1.1.3 MIC (measurement, instrumentation and control) (1)

1.2 1.2.1 Acetylene√

1.2.2 Hydrogen, fuel and combustible gasses containing more than 30% hydrogen or gasses of equivalent hazard.

1.2.3 Ethyl and ethylene√ or gasses of equivalent hazard.

1.2.4 Flour dust, √ grain, wood, plastic and chemicals.
(4 × 1) (4)

1.3
- Calibrating measurement and control instruments
- Analysing electronic circuits
- Fault-finding instruments
- Testing instruments
- Repairing instruments
- Installing instruments
- Maintaining instruments (Any 3 × 1) (3)

[15]

QUESTION 2

2.1 2.1.1 True
2.1.2 True
2.1.3 True
2.1.4 True
2.1.5 True

(5 × 1) (5)

2.2 The balanced actuator is balanced,✓ because the piston pressure areas are equal in chambers A and B.✓ In other words, the same force is exerted in both chambers.✓ (3)

2.3 2.3.1
- Reactive maintenance
  Maintenance that takes place after something has happened, for example, after a valve has stopped working.
- Preventive maintenance
  Is maintenance that is based on a timeline and history.
- Predictive maintenance
  Is when action is taken based on input from instruments and diagnostic tests. (3 × 2) (6)

2.3.2
- Lubrication
- Replacement of the packaging
- Replacement of the seals and gaskets
- Replacement or servicing of worn inner components. (4)

2.3.3
- Fault detection✓
  You monitor a valve and check it until a fault occurs.✓✓
- Fault determination✓
  You inspect the valve until the cause of the fault is found.✓✓
- Corrective action✓
  Involves solving the problem✓✓
- Validation✓
  Is when the valve is installed and run to determine whether it is functional.✓✓ (12)

2.4
- Electric actuator problem
- Hydraulic actuator problem
- Pneumatic actuator problem (3)

2.5
- Are there vapours or mist spraying or flowing from the valve?
- Is there any fluid dripping or leaking from the valve?
- Are there any strange smells that are not usually there?
- Does the valve make any strange noises?
- Is there excessive vibration? (5)

[38]

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QUESTION 3

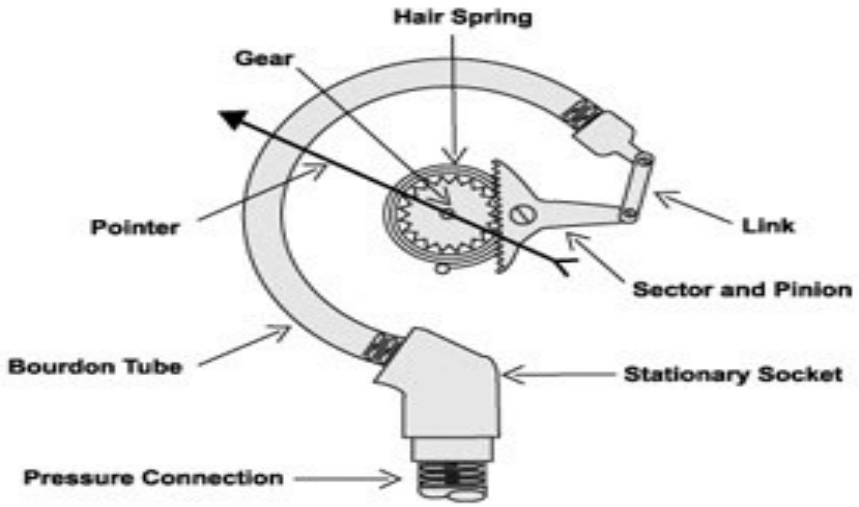
3.1 3.1.1
- Weather studies
- Gas analyses
- Atmospheric surveys
- To research the atmospheres of the planets (Any 3 × 1) (3)

3.1.2 Inclined manometers are useful for measuring pressures √that are very low√ because they are very sensitive to changes in pressure√. (3)

3.1.3
- They cannot be conveniently used for large differences in pressure.
- Certain liquids are not suitable to use in manometers because these do not have well-defined menisci.
- Tension on the surface of the liquid may cause errors because of capillary rise.
- They have slow responses. This makes them unsuitable devices for measuring pressure that fluctuates. (4 × 2) (8)

3.1.4
- Remove the fitting.
- Drain the fluid.
- Rinse the unit with soap and water.
- Replace the O-rings and apply a thin coat of petroleum jelly to ensure a good seal.
- Replace the fluid and return the fittings. (5)

3.2



4 marks for the correctness of the drawing✓✓✓✓
½ Mark for any correct label.√√√√√√√ (8)

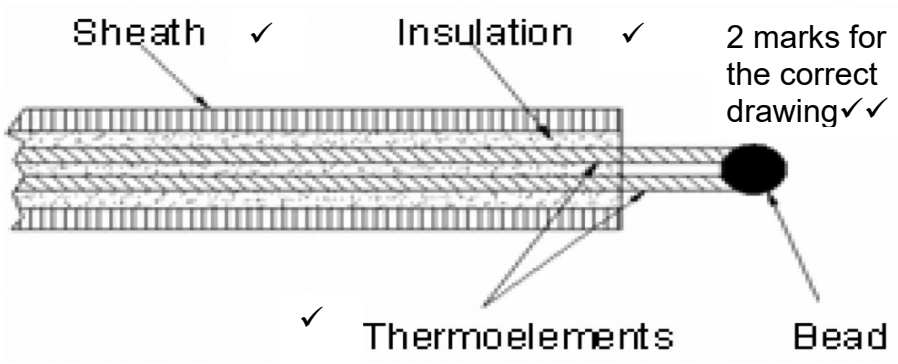
3.3 Charles discovered that at a constant pressure (P),✓ the volume (V) will increase of a given amount of gas (n)✓ if the temperature (T) increases.✓
The volume of a given amount gas is directly proportionate to its temperature if the pressure does not change. (3)
[30]

QUESTION 4

4.1 4.1.1 Generally, when temperature increases, metals expand✓ and when temperature decreases, metals shrink.✓ You can measure their resistance to increase and decrease in temperature.✓ In this way, some metals may convert the changes in temperature to voltage signals.✓ (4)

4.1.2 It is the measure of temperature. (2)

4.2 4.2.1



(5)

4.2.2
- Disconnect the wires from the terminal block to prevent twisting or shorting.
- Insert the thermocouple into the process hole or opening. Tighten the thermocouple in place by turning the thermocouple into the threaded connection.
- Reconnect the wires to the terminal block.
- Connect the thermocouple to a gauge or dial according to the manufacturer's specifications. (5)

4.2.3 Because of the way thermocouples are manufactured, no zero checking✓✓ or calibration can be done since a thermocouple is a sealed unit.✓✓ You should, however, verify that the thermocouple is working by comparing its reading to the temperature reading of another gauge in the area.✓✓ (6)

4.2.4 Install it in its appropriate place;✓ give it a few moments to adjust to the temperature✓ and check to see whether it has a reading on the externally installed gauge.✓ (3)

4.3
- Never apply more than 30 V between any two terminals, or between any terminal and earth/ ground.
- Make sure the battery door is closed and latched before you operate the calibrator.
- Remove the test leads from the calibrator before you open the battery door.
- Do not operate the calibrator if its damaged.
- Do not operate the calibrator around explosive gas, vapour or dust.

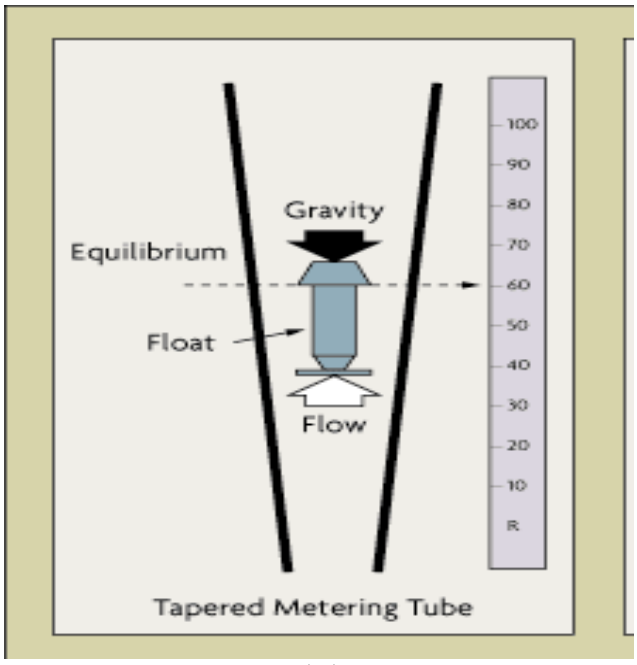
(5)
[30]

QUESTION 5

5.1 Energy = potential energy + kinetic energy
Energy = (pgh) + ($1/2pv^2$) + P✓
Energy = (1000 x 9.81 x 5) + (1/2 x 1000 x 5^2) + 150✓✓
Energy = 49050 + 12500 + 150✓
Energy = 61700 J✓✓ (6)

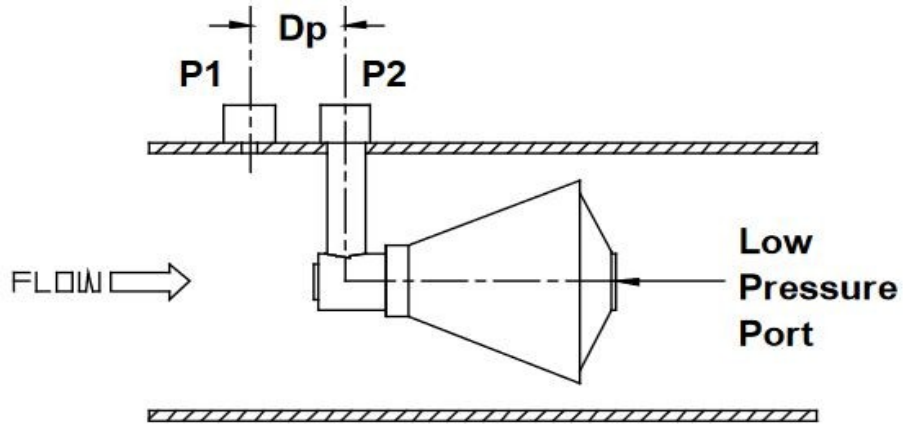
5.2
- There may be particles in the flow path.
- Zero drift
- There may be moisture in the gas.
- Obstruction in the flow path
- Changing pressure
- Not enough pressure
- Not enough power or changing power
- The flow may be too high.
- Output load resistance is too low.
- Too low or too high ambient temperature
- Temperature of gas too high or too low (Any 5 × 1) (5)

5.3 5.3.1



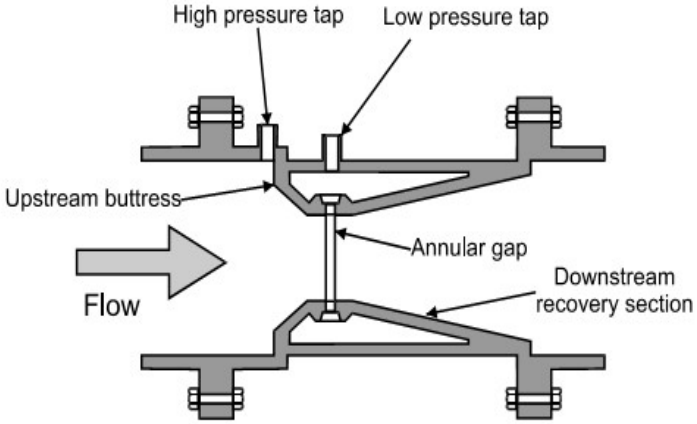
2 Marks for drawing✓✓
1 Mark for each correct label✓✓✓

5.3.2



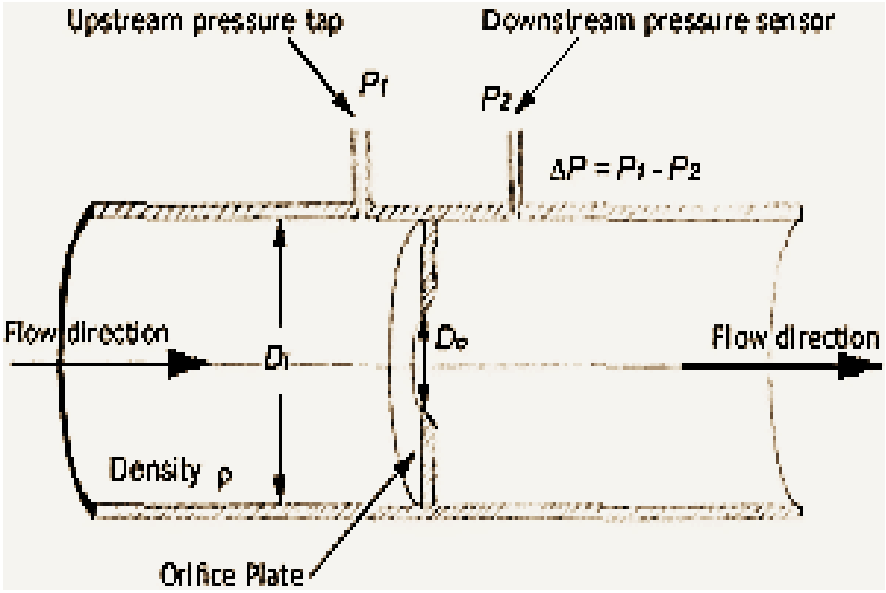
2 Marks for drawing✓✓
1 Mark for each correct label✓✓✓

5.3.3



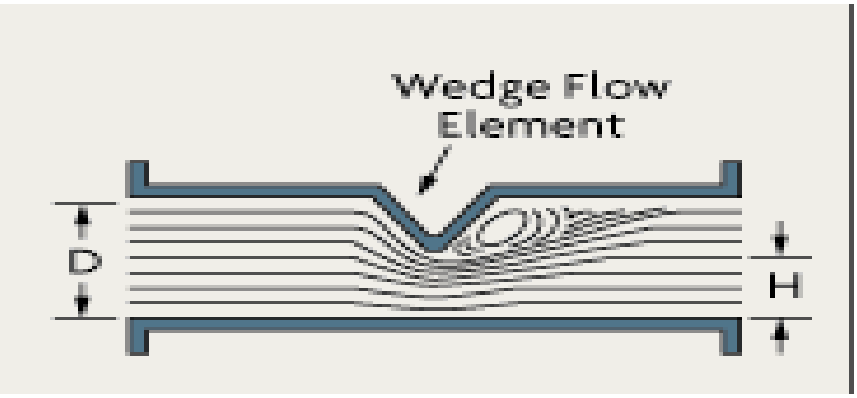
2 Marks for drawing✓✓
1 Mark for each correct label✓✓✓

5.3.4



2 Marks for drawing✓✓
½ Mark for each correct label√√√

5.3.5



2 Marks for drawing✓✓
1 Mark for each correct label✓✓✓

(5 × 5) (25)

5.4 Turbulence (1)

[37]

TOTAL: 150